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Footprint structures due to resonant magnetic perturbations in DIII-D

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Numerical modeling of the typical footprint structures on the target plates of a divertor tokamak is presented. In the tokamak DIII-D [J. L. Luxon, Nucl. Fusion **42**, 614 (2002)] toroidal mode number $n=3$ resonant magnetic perturbations are responsible for characteristic footprint stripes. The numerics can resolve substructures within each footprint stripe, which are related to the internal magnetic topology. It is shown that the footprint structures on the inner target plate can be predicted by the unstable manifolds of the separatrix and the $q=4$ resonant surface. By their intersection with the divertor target plate the unstable manifolds form the footprint boundary and substructures within. Based on the manifold analysis, the boundaries and interior structures of the footprints are explained. A direct connection of all magnetic resonances inside the stochastic plasma volume to the target plates is verified. © *2009 American Institute of Physics*. [DOI: 10.1063/1.3099053]

I. INTRODUCTION

Understanding particle and heat escape mechanisms from open chaotic systems is a central problem in many branches of plasma physics. In magnetic fusion devices, such as divertor tokamaks, the protection of plasma facing wall components from heat and particle overloads is a crucial issue to ensure the operability and desired lifetime of the device. Instabilities at the plasma edge, so called edge localized modes (ELMs),[1] go along with the expulsion of high heat loads which threaten the wall integrity. One of the standard operational scenarios for the next generation tokamak ITER (Ref. 2) will be the type-I ELMing *H*-mode.[3] Power extrapolations for ELMs in ITER show that the plasma facing wall components may suffer from fast erosion or melting.[4]

The use of edge stochastization is a possible solution to control the plasma-wall-interaction.[5] Stochastization of the plasma edge may influence the heat and particle flows to the wall.[1,6,7] In DIII-D[8] resonant magnetic perturbations (RMPs) have successfully been used in ELM mitigation.[1] Also type-I ELM control was recently achieved by RMP in JET.[9] Motivated by these results, a flexible set of RMP coils was added to the ITER design[10,11] in the recent design review process.

Theory indicates that the ELMs are eliminated by the peeling-ballooning instability threshold. It was found[1,5] that simple stochastic diffusion theory overestimates the electron thermal transport determined from the experimental data. The experimental results indicate that stochastic transport theory and rotational shielding theory are incomplete when applied to highly rotating collisionless pedestal plasmas. The particle and energy transport in the tokamak edge transport barrier were then analyzed in more detail in the presence of magnetic field perturbations from external resonant coils. The observed reduction of density without decrease of electron temperature[9] was successfully explained by generation of charged particle flow along perturbed field lines.[12,13]

The magnetic topology of an ergodized edge shows two main regions. The ergodic zone[14] consists of remanent island chains, related to resonances, surrounded by stochastic field lines which diffuse radially. The laminar zone,[14,15] connecting the pedestal plasma with the divertor target plates, is characterized by very short field lines. The transport properties of the laminar zone have been investigated in great detail.[15,16] Especially so called laminar flux tubes[17] are suspected to be a major sink for heat and particles. Results from TEXTOR (Ref. 14) show that the ergodization of the plasma is well described by the vacuum field approximation.[18] However shielding effects due to high plasma rotation could play a crucial role in other devices. A comparison of the predicted heat and particle flux patterns with actual measurements can be used to reveal differences and agreements to the vacuum field approach.

In order to analyze and classify the spatial structures of heat and particle flux patterns, the concept of magnetic footprints[19–21] (intersection structure of field lines on the divertor target plates) is very helpful. We combine it with an analysis of the stable and unstable manifolds of hyperbolic periodic points[22–25] of field lines. The unstable manifold of a hyperbolic periodic point is defined as the set of points in the Poincaré section (given by the intersections of field lines with this section) which converge toward the hyperbolic point under the equations of motion for the field lines for the toroidal angle $\varphi=2\pi k$ and $k\rightarrow\infty$ with $k\in\mathbb{Z}$. The stable manifold is the unstable manifold for the reverse toroidal direction $k\rightarrow-\infty$. Research at the limiter tokamak TEXTOR led to a selection rule,[26] which connects the magnetic footprints to actual heat flux measurements by using stable and unstable manifolds. Good agreement was found with the observed patterns.[27] It is now interesting to investigate whether that concept also applies to other machines.

Here we focus on the poloidally diverted tokamak DIII-D with RMP fields due to a nonaxisymmetric perturbation coil located inside the DIII-D vacuum vessel near the plasma edge, known as the I-coil.[1] The main purpose of the present paper is to describe and explain the generic footprint structures on the divertor target induced by RMP. Especially the link between the internal resonant magnetic flux surfaces,

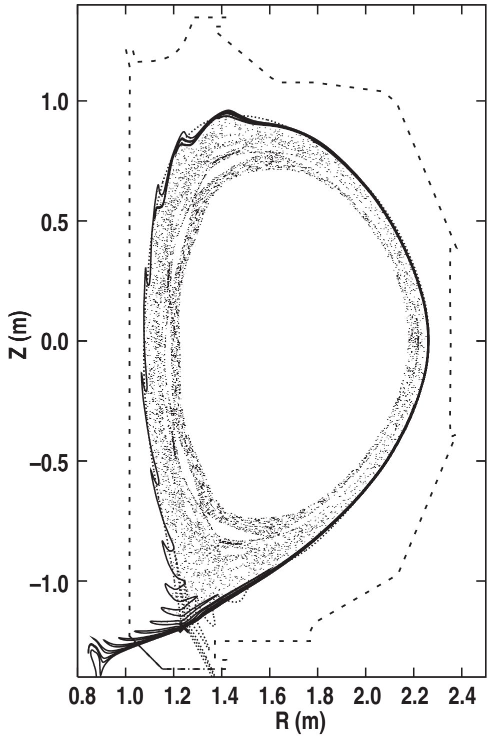

FIG. 1. Stochastic magnetic field for an ELM suppressed H-mode plasma with $q_{95}=3.55$ (Shot 126006 at 3000 ms) and the I-coil only configured for a $n=3$ toroidal mode number perturbation. The I-coil current is about 4.3 kA. The figure is for a toroidal angle $\varphi=0$. Lower hyperbolic point (cross) and respective unstable manifold (solid line) are presented. The stable manifold that makes up the outer divertor leg intersecting the outer target plate is shown by the dotted line. The DIII-D vacuum vessel boundary is given by the dashed line. The inner and outer targets are depicted by the solid and dotted-dashed part of the wall, respectively. Note that the $q=4$ unstable manifold of Fig. 2 (red line in Fig. 2) is not presented in this figure.

defined by rational q_{95} values, and the footprint structures is visualized by the manifold analysis. In Sec. II the structures of the unstable separatrix manifold and an unstable manifold related to an internal resonance are presented and discussed. The footprint structures are evaluated in detail for this case in Sec. III. The connection of the footprints with the internal topology is shown and discussed. The last section summarizes the conclusions.

II. UNSTABLE MANIFOLDS OF THE SEPARATRIX AND INTERNAL RESONANCES

Although the results shown here are derived using the I-coil only, different sets of coil arrangements (such as the C-coil used for correcting field errors in DIII-D) and uncorrected field errors can cause RMP fields with various toroidal mode numbers. Such fields create structures similar to those shown here. Therefore it is assumed that the results and principles derived here are directly applicable to any kind of RMP in poloidally diverted tokamaks, including ITER. Especially for ITER the prediction of position, size and structure of heat flux patterns on the divertor targets is of great interest.

Figure 1 shows a typical stochastic magnetic field created by the I-coils for a lower single null plasma calculated with an invariant manifold code (MAFOT) that has been coupled to the TRIP3D (Ref. 28) vacuum field line integration code. The combination is called the TRIP3D-MAFOT code. The MAFOT code uses the procedure outlined in Ref. 22 to calculate the structure of the stable and unstable invariant manifolds. Discharge data are passed to it from the TRIP3D code. A cylindrical coordinate system (R,φ,Z) is used with R being the major radius. The Poincaré section is determined by a constant toroidal angle; we have chosen $\varphi=0$ here. The magnetic equilibrium is reconstructed by using experimental data.[29] The perturbation is the vacuum field modeled from the actual coil currents of the shot at the respective time. Due to the RMP, the separatrix splits into the stable and unstable manifolds.[19] The latter, with respect to positive toroidal direction, is shown in the figure by the solid line. As can be seen, the unstable manifold intersects several times with the inner divertor target plate centered at $R=1.07$ m, $Z=-1.33$ m. Since the stable and unstable manifolds act as separatrices, field lines inside the plasma volume are allowed to connect to the inner target plate only through the loops of the stable and unstable manifolds that intersect the surface of the wall. So we expect the unstable manifold to be the main boundary of the footprints on the inner target plate. Note that the same is valid for the stable manifold (when following field lines in negative toroidal direction). A stable manifold is shown in Fig. 1 by the dotted line. It hits the outer target plate located near $R=1.36$ m.

Each resonant mode of the magnetic perturbation spectrum creates an island chain. The latter is located at the respective radial position where the q-profile equals the ratio of the poloidal and toroidal mode number. The radial overlapping of different island chains forms the stochastic regions. Intact island chains have separatrices as well.[19] With increasing perturbations these separatrices split into stable and unstable manifolds. The paths of the latter appear in the Poincaré plot as loops moving back and forth in the vicinities of the islands, thereby densely filling the entire stochastic region. Note that each stable/unstable manifold intersects infinite times with any other unstable/stable manifold within the stochastic region. In an open chaotic system all manifolds are finally connecting to the wall. The pathways of stable and unstable manifolds determine the preferred heat and particle flow channel induced by the RMP magnetic topology.[26] In the present paper, the structures of the connection length patterns on the divertor plates, which match the heat and particle flow patterns,[27] will be related to the intersections of the manifold pathways with the divertor plates.

The unstable manifolds of the internal resonant surfaces, as far as they are open to the stochastic region, are embedded within the loops of the unstable separatrix manifold. This can be seen in Fig. 2. The latter shows an enlargement of the part of Fig. 1 close to the inner target plate. The unstable separatrix manifold of Fig. 1 is given by the black line. The red line is an unstable manifold associated with a magnetic island on the $q=4$ resonant surface with a poloidal mode number $m=12$ and a toroidal mode number $n=3$. This island is located on the $\psi_N=0.978$ normalized poloidal flux surface inside the separatrix with a width of $\Delta\psi_N=0.024$. These data follow from the TRIP3D-MAFOT vacuum field line integration code. The island chain is located at approximately the point

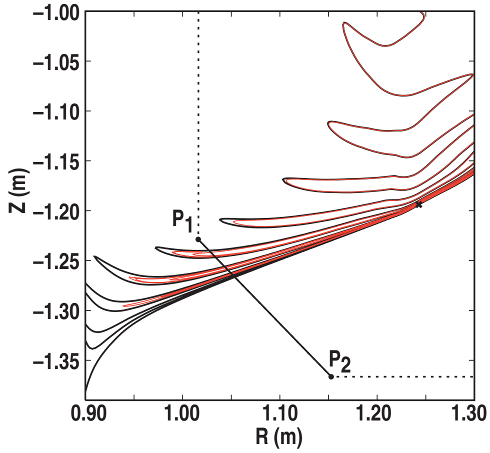

FIG. 2. (Color) Magnification of the target plate region of Fig. 1. The unstable manifold of the separatrix (black line) and an unstable manifold of the $q=4$ resonant surface (red line) are shown. P_1 and P_2 indicate the end points of the target plate (solid black line segment along the dashed wall).

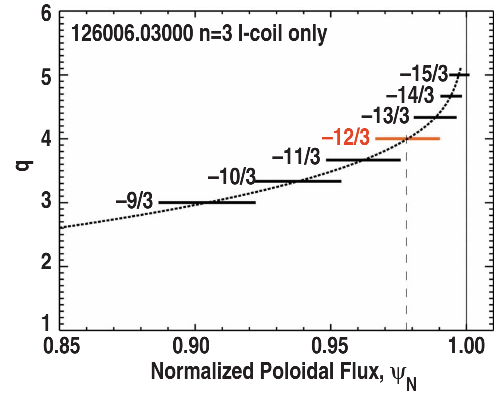

FIG. 3. (Color online) Calculation of the locations and widths of $n=3$ magnetic islands produced by the I-coil that create the stochastic layer shown in Fig. 1. An unstable manifold from the $m,n=12,3$ island located on the $q=4$ resonant surface is shown by the red line in Fig. 2.

where the plasma pressure gradient is a maximum, at about one-half the distance between the separatrix and the top of the H-mode plasma pedestal region It fills most of this high gradient region. Therefore we focus on this particular island chain. Part of the unstable manifold of the $m,n=12,3$ island chain intersects with the wall in several loops within the loops of the unstable separatrix manifold. Thereby internal structures are formed. Thus we expect the footprints to show a nested structure. The structure is related to field lines generated by the different manifolds, caused by the RMP mode spectrum, of resonances to the different q_{95} flux surfaces, which is not the case in absence of RMP. The $n=3$ island widths and locations are shown in Fig. 3. They are calculated by making use of the EFIT equilibrium data and vacuum RMP perturbations.

III. FOOTPRINTS AND THEIR RELATION TO THE STABLE AND UNSTABLE MANIFOLDS

In this section we first calculate the number of toroidal rotations necessary to connect a field line, which starts on the inner target plate, with the outer target plate. We start at different positions by introducing a grid on the inner target plate. Having found the number of rotations until a field line hits the outer target plate, we code the number of rotations by color. Thereby, we obtain the colored contour plot shown in Fig. 4(a). Figure 4(b) shows the footprint structures on the outer target. In DIII-D the diagnostic possibilities on the outer target are limited in high triangularity shaped plasmas (like the shot considered here). Camera views of the outer target are partly blocked by the exhaust vent. So, in the following we focus on the inner target.

We use a special coordinate system for the surface of the inner target plate. The dimensionless length $t_{12}\in[0,1]$ parametrizes the position on the target plate in the (R,Z) plane (see Fig. 2), and φ is the toroidal angle. The target plate has the end points $P_1=(1.0161,-1.228\,84)$ and $P_2=(1.152\,85,-1.3664)$ (coordinates are in meters), as shown in Fig. 2. For a point P on the target plate we define in the (R,Z) plane,

$$t_{12}=\frac{|P-P_1|}{|P_2-P_1|}. \quad (1)$$

Figure 4(a) shows that areas of different connection lengths form characteristic patterns. We recognize three different distinct stripes, which periodically continue in φ with increasing t_{12}. The number of different stripes is given by the toroidal mode number n of the RMP. Here, only the I-coil is used with $n=3$. The stripes are getting thinner and closer together with increasing t_{12}. With increasing t_{12} they

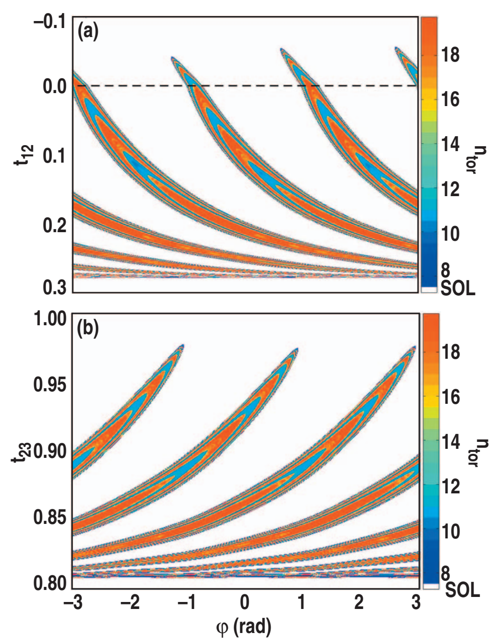

FIG. 4. (Color) (a) Magnetic footprints on the inner target plate. The color indicates the number of toroidal rotations necessary to connect to the outer target (n_{tor}). The SOL is given by the white areas. A dimensionless length parameter t_{12}, which indicates to position on the target plate in the (R,Z) plane, is plotted vs the toroidal angle $\varphi\in[-\pi,\pi]$. The dashed line indicates the connection of the inner target plate with the vertical inner wall. Negative t_{12} values specify positions on the inner wall. (b) Same as (a) but for the outer target. The dimensionless length t_{23} is directly related to the radial coordinate R. We have $R=1.152\ \mathrm{m}+0.219t_{23}\ \mathrm{m}$.

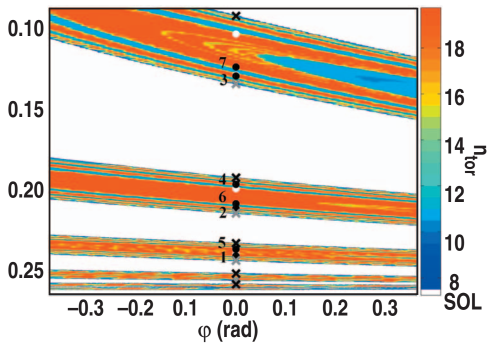

FIG. 5. (Color) Magnification of Fig. 4(a) around $\varphi=0$. Intersections of the unstable separatrix manifold are shown by crosses (black: outgoing, white: incoming). Intersections of an unstable manifold originating from a $q=4(m,n=12,3)$ resonant island chain are shown by dots (colors accordingly, numbers according to Fig. 6).

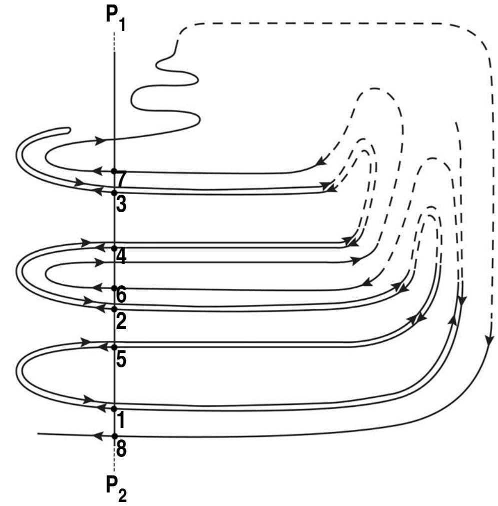

FIG. 6. Sketch of the intersection scheme of the unstable manifold of the islands at the $q=4$ surface, shown by the red line in Fig. 2, with the inner target plate. The intersections are numbered in sequence as they occur. The dashed segments indicate the (in reality highly complicated structure of the) unstable manifold inside the plasma. The sizes of the loops are not true to scale.

approach the position of the unperturbed separatrix at $t_{12}\approx 0.275$. Within each stripe one finds substructures which appear as looplike bands. Areas of large and short connection lengths alternate when crossing through a stripe. Field lines belonging to the areas of short connection lengths connect the two target plates without penetrating into the stochastic plasma region.[30] In the following we focus on the large connection length bands, which are fractal in nature.

To understand the fractal structure we determine the intersection points of the unstable manifolds shown in Figs. 1 and 2 with the inner target plate and mark them in the footprint pattern, Fig. 4(a). The result is shown in Fig. 5. An enlargement of the footprints around $\varphi=0$ is shown. The crosses mark the intersections of the unstable separatrix manifold; their colors indicate whether the unstable manifold leaves the vessel (black) or reenters the vessel (gray) through the target plate. As expected, the unstable separatrix manifold acts as boundary for the footprint stripes. It strikes the target plate only at the edges of the stripes. All field lines with large connection lengths, coming from inside the plasma, strike the target inside the loops of the unstable separatrix manifold. Field lines outside of the stripes are from the scrape-off layer (SOL) which never enter the plasma and have very short connection lengths. So, the shape and position of the stripes on the target plates follow from the stable and unstable separatrix manifolds for the outer and inner target plates, respectively.

The black and white dots mark the respective intersections of an outgoing and incoming unstable manifold originating from a $q=4$ $(m,n=12,3)$ internal resonant surface. The unstable manifolds of internal resonant surfaces strike the target in a structure of long connection lengths, visualized by red bands, inside the stripes and thereby create the fractal substructures of the footprints. The bands closer to the stripe boundary are related to resonant surfaces closer to the separatrix. An unstable manifold can strike at several bands since it follows a highly complicated path on its way to the target plate.

Let us elucidate the behavior for a typical example. The $q=4$ resonant surface consists of four sets of hyperbolic points. Each set intersects the poloidal plane three times before returning to the initial poloidal position. Each set of hyperbolic points has two stable and two unstable directions. So, the $q=4$ surface has eight different unstable manifolds. Since they are not allowed to intersect, they are convoluted in a quite complex manner resulting in a highly complicated path. Besides the back and forth going oscillations in poloidal direction the pathway spirals radially toward the separatrix and finally hits the target plate.

As shown in Fig. 5 the $q=4$ unstable manifold intersects with each stripe several times. The intersection scheme is sketched in Fig. 6. Note that $t=0$ corresponds to the position P_1 on the target plate. So the upper stripe in Fig. 5 corresponds to the upper loop with the intersection numbers 3 and 7 in the sketch. The numbers in both figures give the sequence of intersections of the unstable manifold with the target. Here, only the outgoing intersections (coming from the plasma interior) are numbered. The direction of the unstable manifold is given by the black arrows along the unstable manifold. The dashed lines in the sketch indicate the (complicated) manifold paths inside the plasma.

In Fig. 5, the first intersection is closest to the unperturbed separatrix at $t_{12}\approx 0.275$. From there subsequent intersections step through each stripe (see numbering). After each reenter, the unstable manifold penetrates into the plasma, as indicated by the dashed lines in Fig. 6. According to Fig. 5 the intersections 6 and 7 are within the second band further inside the stripe. This indicates that the unstable manifold is convoluted with unstable manifolds of resonant surfaces below $q=4$. Let us look on the path of the unstable manifold in more detail. On its path the unstable manifold penetrates radially deeper inside the perturbed plasma volume before getting further outside again. This seems to contradict the fact that manifolds do not intersect. However, we should have in mind that the unstable manifold has not yet completed its first full poloidal turn. After intersection 7 the unstable manifold completes a full poloidal turn and ends up very close to the separatrix. In that way the unstable manifold is allowed to spiral radially to the outside. For computational reasons the path of the unstable manifold could not be followed any further (intersection 8 is not show in Fig. 5). However the scheme continues in the same manner until the unstable manifold becomes infinitesimally close to the separatrix.

IV. CONCLUSION

The essential conclusion drawn from the theory presented here is that the interior structure of the footprint stripes relate directly to the resonances on rational mode surfaces inside the perturbed plasma volume. Unlike TEXTOR, where the last resonance in front of the wall determines the wall patterns,[26] in DIII-D all of the resonances in the outer part of the stochastic region are directly connected to the target plates and therefore contribute to the heat flux incident on the target. Each of the fractal bands seen in the footprint substructures on the inner target is a composite of the spiraling unstable manifolds from resonant surface space across the edge of the plasma, as shown in Fig. 3. The further the band is inside the stripe, the further the resonance is inside the plasma, and vice versa. Increasing the amplitude of the perturbing field, stable and unstable manifolds from deeper resonances connect to the target plates. An increasingly more complex internal structure in the footprint patterns is created since the field lines follow the directions of more and more manifolds before they finally reach the target plates. The boundaries of the footprint stripes are determined by the stable and unstable separatrix manifolds. The heat load is expected to be proportional to the area of the stripes and will increase with the perturbation depth characterizing the substructures. All results are valid either for the inner target plate and unstable manifolds or for the outer target plate and stable manifolds. The footprints on both target plates are alike. Note that a stable manifold is the respective unstable one for the reverse toroidal direction and therefore shows similar structures on the outer target plate.

The present theory predicts, in addition to locations and areas of the footprint stripes, a fine structure which, however, at present is beyond experimental evidence from target measurements. Observability of the fine structure is related to the importance of perpendicular diffusion processes as well as screening of the vacuum RMP by induced plasma currents. Thus more detailed experiments on heat flux patterns will lead to a better understanding of the dominating processes.

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[1] T. E. Evans, R. A. Moyer, P. R. Thomas, J. G. Watkins, T. H. Osborne, J. A. Boedo, E. J. Doyle, M. E. Fenstermacher, K. H. Finken, R. J. Groebner, M. Groth, J. H. Harris, R. J. LaHaye, C. J. Lasnier, S. Masuzaki, N. Ohyabu, D. G. Pretty, T. L. Rhodes, H. Reimerdes, D. L. Rudakov, M. J. Schaffer, G. Wang, and L. Zeng, Phys. Rev. Lett. **92**, 235003 (2004).

[2] ITER Physics Basis Editors, ITER Physics Expert Group Chairs and Co-Chairs, and ITER Joint Central Team and Physics Integration Unit, Nucl. Fusion **39**, 2137 (1999).

[3] J. W. Connor, A. Kirk, and H. R. Wilson, AIP Conf. Proc. **1013**, 174 (2008).

[4] A. Loarte, G. Saibene, R. Sartori, D. Campbell, M. Becoulet, L. Horton, T. Eich, A. Herrmann, G. Matthews, N. Asakura, A. Chankin, A. Leonard, G. Porter, G. Federici, G. Janeschitz, M. Shimada, and M. Sugihara, Plasma Phys. Controlled Fusion **45**, 1549 (2003).

[5] T. E. Evans, R. A. Moyer, K. H. Burrell, M. E. Fenstermacher, I. Joseph, A. W. Leonard, T. H. Osborne, G. D. Porter, M. J. Schaffer, P. B. Snyder, P. R. Thomas, W. P. West, and J. G. Watkins, Nat. Phys. **2**, 419 (2006).

[6] P. Ghendrih, A. Grossman, and H. Capes, Plasma Phys. Controlled Fusion **38**, 1653 (1996).

[7] R. Balescu, *Aspects of Anomalous Transport in Plasmas* (Institute of Physics, Bristol, 2005).

[8] J. L. Luxon, Nucl. Fusion **42**, 614 (2002).

[9] Y. Liang, H. R. Koslowski, P. R. Thomas, E. Nardon, B. Alper, P. Andrew, Y. Andrew, G. Arnoux, Y. Baranov, M. Bécoulet, M. Beurskens, T. Biewer, M. Bigi, K. Crombe, E. D. L. Luna, P. de Vries, W. Fundamenski, S. Gerasimov, C. Giroud, M. P. Gryaznevich, N. Hawkes, S. Hotchin, D. Howell, S. Jachmich, V. Kiptily, L. Moreira, V. Parail, S. D. Pinches, E. Rachlew, and O. Zimmermann, Phys. Rev. Lett. **98**, 265004 (2007).

[10] M. Bécoulet, E. Nardon, G. Huysmans, W. Zwingmann, P. Thomas, M. Lipa, R. Moyer, T. Evans, V. Chuyanov, Y. Gribov, A. Polevoi, G. Vayakis, G. Federici, G. Saibene, A. Portone, A. Loarte, C. Doebert, C. Gimblett, J. Hastie, and V. Parail, Nucl. Fusion **48**, 024003 (2008).

[11] M. J. Schaffer, J. E. Menard, M. P. Aldan, J. M. Bialek, T. E. Evans, and R. A. Moyer, Nucl. Fusion **48**, 024004 (2008).

[12] M. Z. Tokar, T. E. Evans, A. Gupta, R. Singh, P. Kaw, and R. C. Wolf, Phys. Rev. Lett. **98**, 095001 (2007).

[13] M. Z. Tokar, A. Gupta, D. Kalupin, and R. Singh, Plasma Phys. Controlled Fusion **49**, 395 (2007).

[14] K. H. Finken, S. S. Abdullaev, A. Kaleck, and G. H. Wolf, Nucl. Fusion **39**, 637 (1999).

[15] T. Eich, D. Reiser, and K. H. Finken, Nucl. Fusion **40**, 1757 (2000).

[16] F. Nguyen, P. Chendrih, and A. Grosman, Nucl. Fusion **37**, 743 (1997).

[17] O. Schmitz, M. W. Jakubowski, H. Frerichs, D. Harting, M. Lehnen, B. Unterberg, S. S. Abduallaev, S. Brezinsek, I. Classen, T. E. Evans, Y. Feng, K. H. Finken, M. Kantor, D. Reiter, U. Samm, B. Schweer, G. Sergienko, G. W. Spakman, M. Tokar, E. Uzgel, and R. C. Wolf, and TEXTOR Team, Nucl. Fusion **48**, 024009 (2008).

[18] M. W. Jakubowski, O. Schmitz, S. S. Abdullaev, S. Brezinsek, K. H. Finken, A. KrŠmer-Flecken, M. Lehnen, U. Samm, K. H. Spatschek, B. Unterberg, R. C. Wolf, and TEXTOR Team, Phys. Rev. Lett. **96**, 035004 (2006).

[19] A. Punjabi, A. Verma, and A. Boozer, Phys. Rev. Lett. **69**, 3322 (1992).

[20] E. C. da Silva, I. L. Caldas, R. L. Viana, and M. A. F. Sanjuán, Phys. Plasmas **9**, 4917 (2002).

[21] S. S. Abdullaev, T. Eich, and K. H. Finken, Phys. Plasmas **8**, 2739 (2001).

[22] E. Nusse and J. A. Yorke, *Dynamics: Numerical Explorations* (Springer, New York, 1998).

[23] A. Wingen, K. H. Spatschek, and S. S. Abdullaev, Contrib. Plasma Phys. **45**, 500 (2005).

[24] T. E. Evans, R. K. Roeder, J. A. Carter, and B. I. Rapoport, Contrib. Plasma Phys. **44**, 235 (2004).

[25] T. E. Evans, R. K. Roeder, J. A. Carter, B. I. Rapoport, M. E. Fenstermacher, and C. J. Lasnier, J. Phys.: Conf. Ser. **7**, 174 (2005).

[26] A. Wingen, M. W. Jakubowski, K. H. Spatschek, S. S. Abdullaev, K. H. Finken, and M. Lehnen, and TEXTOR Team, Phys. Plasmas **14**, 042502 (2007).

[27] M. W. Jakubowski, S. S. Abdullaev, K. H. Finken, M. Lehnen, and TEXTOR Team, J. Nucl. Mater. **337–339**, 176 (2005).

[28] T. E. Evans, R. A. Moyer, and P. Monat, Phys. Plasmas **9**, 4957 (2002).

[29] L. Lao, H. St. John, R. D. Stambaugh, A. G. Kellman, and W. Pfeiffer, Nucl. Fusion **25**, 1611 (1985).

[30] A. Wingen, T. E. Evans, and K. H. Spatschek, "High resolution numerical studies of separatrix splitting due to non-axisymmetric perturbation in DIII-D," Nucl. Fusion (in press).